

In the following exercises, all coordinates and components are given with respect to a right-oriented orthonormal frame $\mathcal{K}$.

6.1. The vertices of a triangle are $A(1, 1)$, $B(4, 1)$ and $C(2, 3)$. Determine the image of the triangle ABC under a rotation by 90° around C followed by an orthogonal reflection in the line AB .

6.2. Consider the affine transformation given in homogeneous coordinates by

$$\phi\left(\begin{bmatrix} x \\ y \\ 1 \end{bmatrix}\right) = \frac{1}{5} \begin{bmatrix} 4 & -3 & 7 \\ 3 & 4 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}.$$

Prove that it is a rotation. Moreover, determine

- a) the cosine of the rotation angle,
- b) the center of rotation,
- c) the inverse ϕ^{-1} .

6.3. Let T be the isometry obtained by applying a rotation with angle $-\frac{\pi}{3}$ around the origin after a translation with vector $(-2, 5)$. Determine the inverse transformation, T^{-1} .

6.4. Let $\text{Rot}_{\theta_1, C_1}$ and $\text{Rot}_{\theta_2, C_2}$ be rotations with angles θ_1, θ_2 and centers C_1, C_2 respectively. Show that the composition of these two rotations is a (possibly trivial) translation if and only if $\theta_1 + \theta_2 = 2k\pi$ for some integer k .

6.5. Consider the line $\ell : x - 2y + 1 = 0$. Give a glide-reflection in the line ℓ which moves the point $(-3, 0)$ to the point $(\frac{9}{5}, \frac{2}{5})$.

6.6. Show that a rotation with angle θ around the origin after a reflection in the x -axis is a reflection in the line $y = \tan(\theta/2)x$.

6.7. Discuss, in dimension 2, the possible isometries obtained by composing a reflection in a line ℓ_1 with a reflection in a line ℓ_2 in terms of the relative positions of the two lines.